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(54) **PLASMA PROCESSING APPARATUS AND
SUBSTRATE PROCESSING METHOD**

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(2013.01); **H01J 37/3244** (2013.01); **H01J**
37/32715 (2013.01); **H01J 2237/327** (2013.01)

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CPC H01J 37/32146; H01J 37/32422; H01J
37/3244; H01J 37/32715; H01J 2237/327
See application file for complete search history.

(56) **References Cited**

U.S. PATENT DOCUMENTS

2022/0122810 A1 * 4/2022 Kubota H01J 37/32091
2022/0384150 A1 * 12/2022 Koshimizu H01J 37/32183
2023/0162946 A1 * 5/2023 Koshimizu H01J 37/32706
315/111.21

(Continued)

FOREIGN PATENT DOCUMENTS

CN 102067279 A * 5/2011 B08B 5/00
JP 2019-169635 A 10/2019

(Continued)

OTHER PUBLICATIONS

International Search Report and Written Opinion mailed on Jun. 7,
2022, received for PCT Application PCT/JP2022/016485, filed on
Mar. 31, 2022, 10 pages including English Translation.

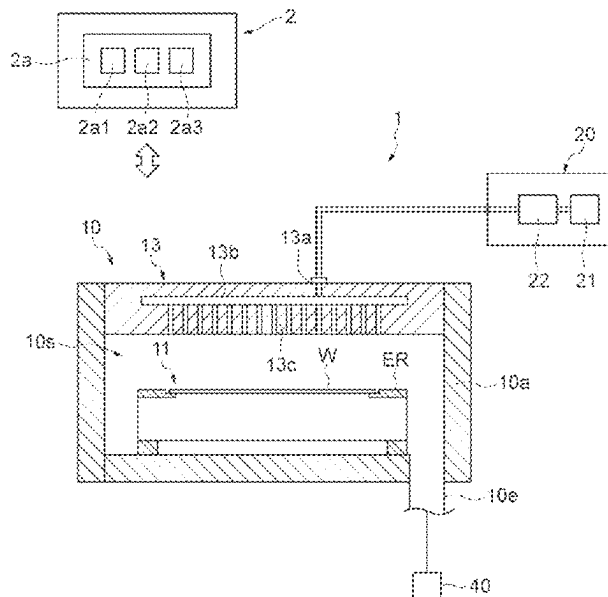
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(57) **ABSTRACT**

A plasma processing apparatus comprises a chamber, a substrate support, a plasma generator, a bias power supply and a chuck power supply. The substrate support includes a base and a dielectric part. The base includes a base member and an electrode. The base member is made of a dielectric or an insulator. The electrode is formed on an upper surface of the base member. The electrode forms an upper surface of the base. The dielectric part provides a supporting surface on which a substrate is placed. The dielectric part extends from the upper surface of the base to the supporting surface and is made of only a dielectric. The bias power supply and the chuck power supply are electrically connected to the electrode of the base.

13 Claims, 12 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

2023/0260766	A1 *	8/2023	Torii	H01J 37/32642
				315/111.21
2024/0062991	A1 *	2/2024	Koshimizu	H01J 37/32706
2025/0046573	A1 *	2/2025	Bin Budiman	H01L 21/3065
2025/0069851	A1 *	2/2025	Koshimizu	H01J 37/32715
2025/0096006	A1 *	3/2025	Sasaki	H01L 21/31116

FOREIGN PATENT DOCUMENTS

JP	2020-115541	A	7/2020
JP	2020-534668	A	11/2020
JP	2020-205444	A	12/2020
WO	2019/060028	A1	3/2019
WO	2020/159674	A1	8/2020
WO	WO-2024162171	A1 *	8/2024

* cited by examiner

FIG. 1

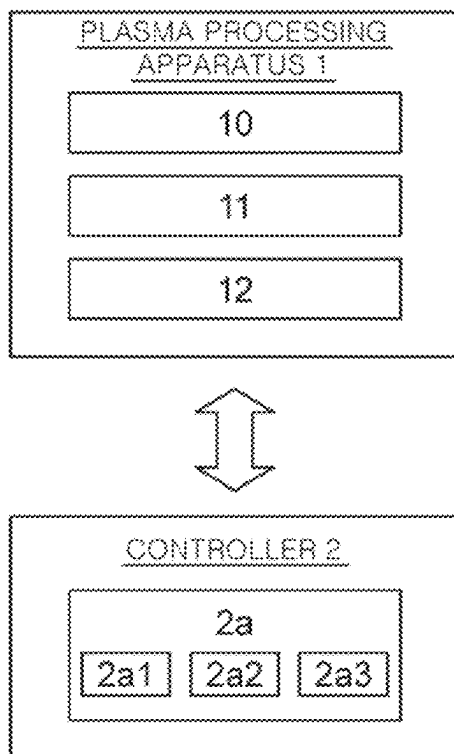


FIG. 2

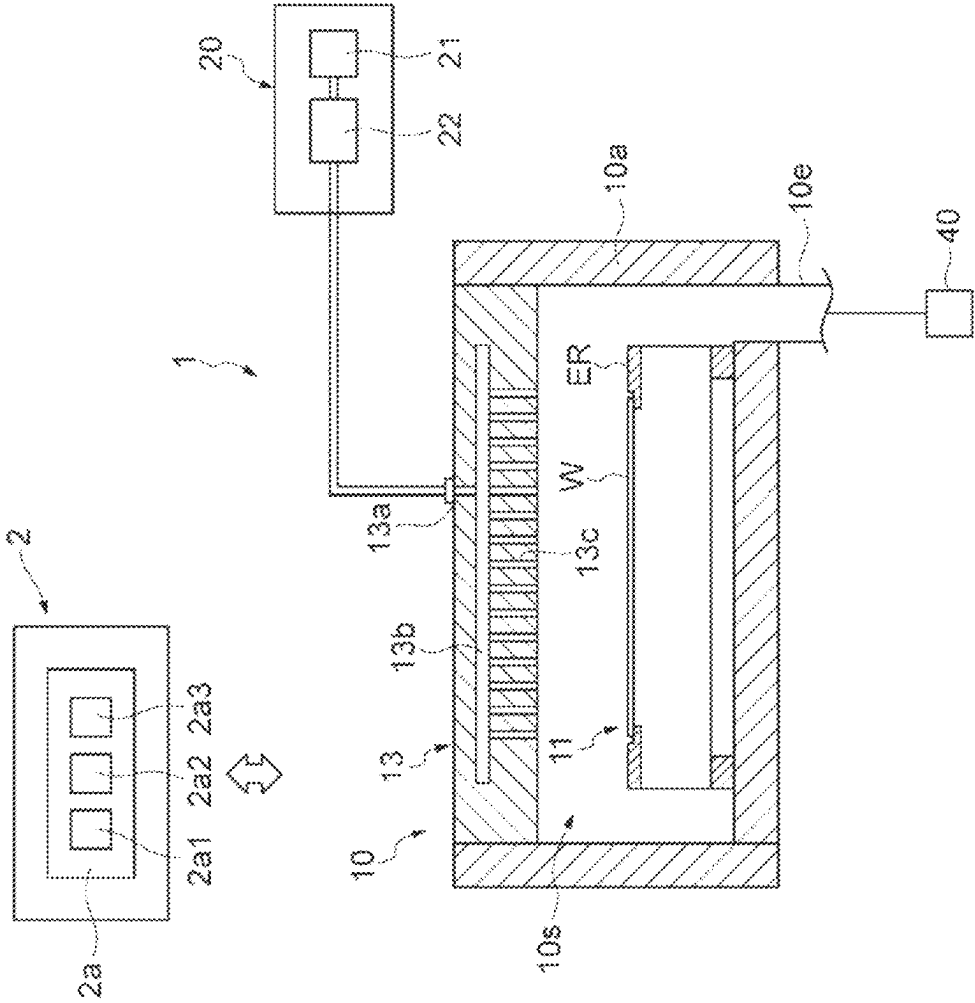


FIG. 3

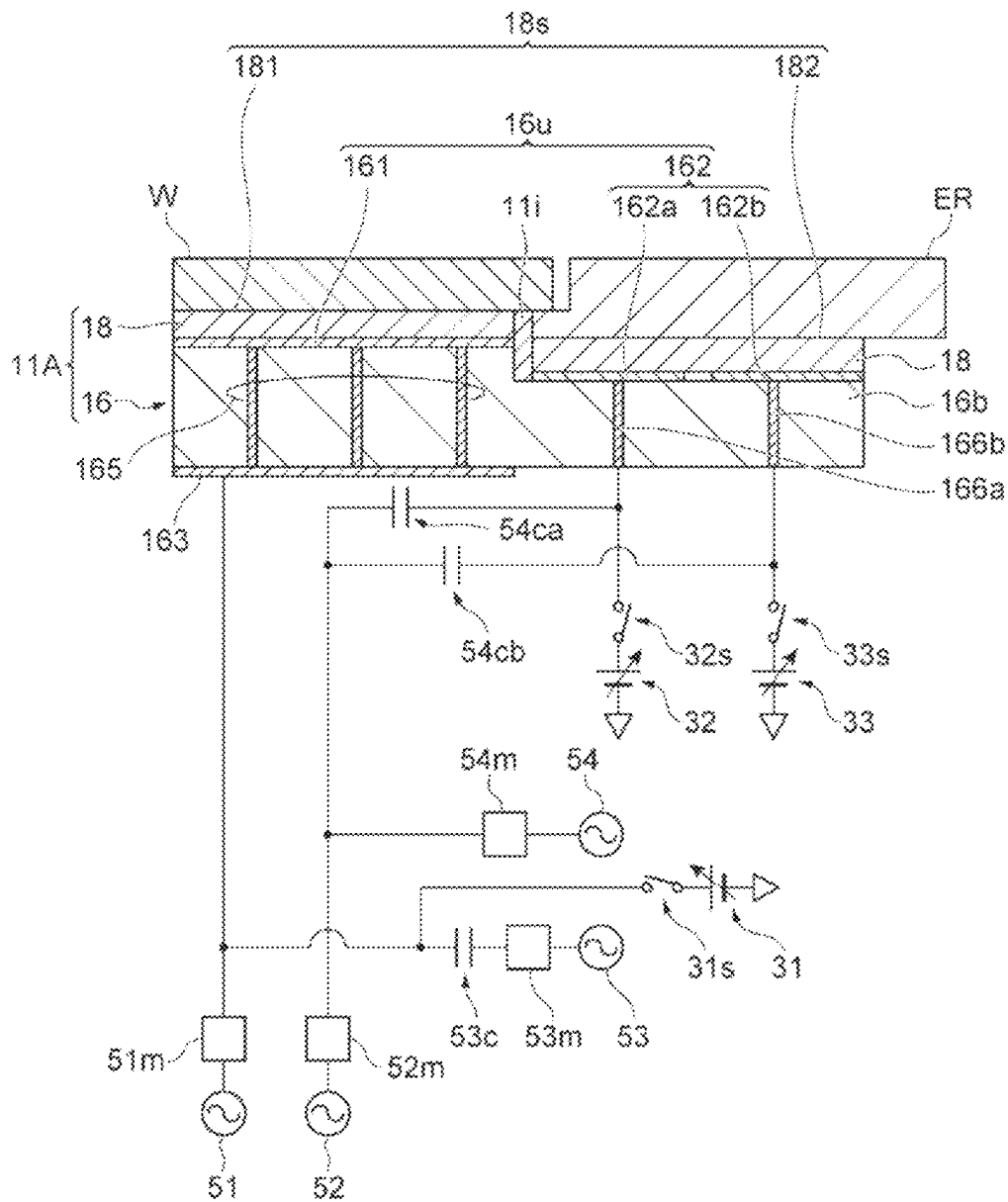


FIG. 4

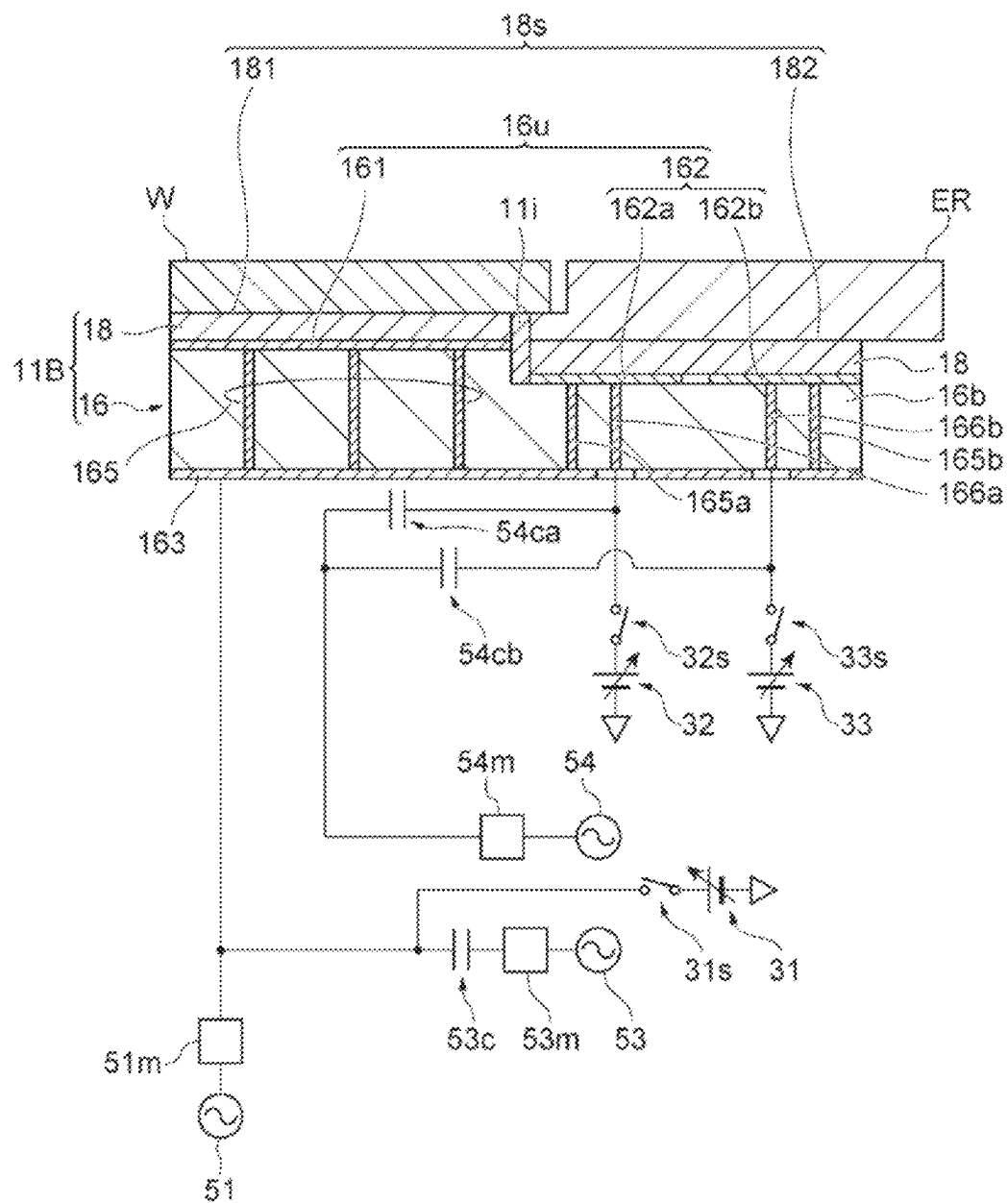


FIG. 5

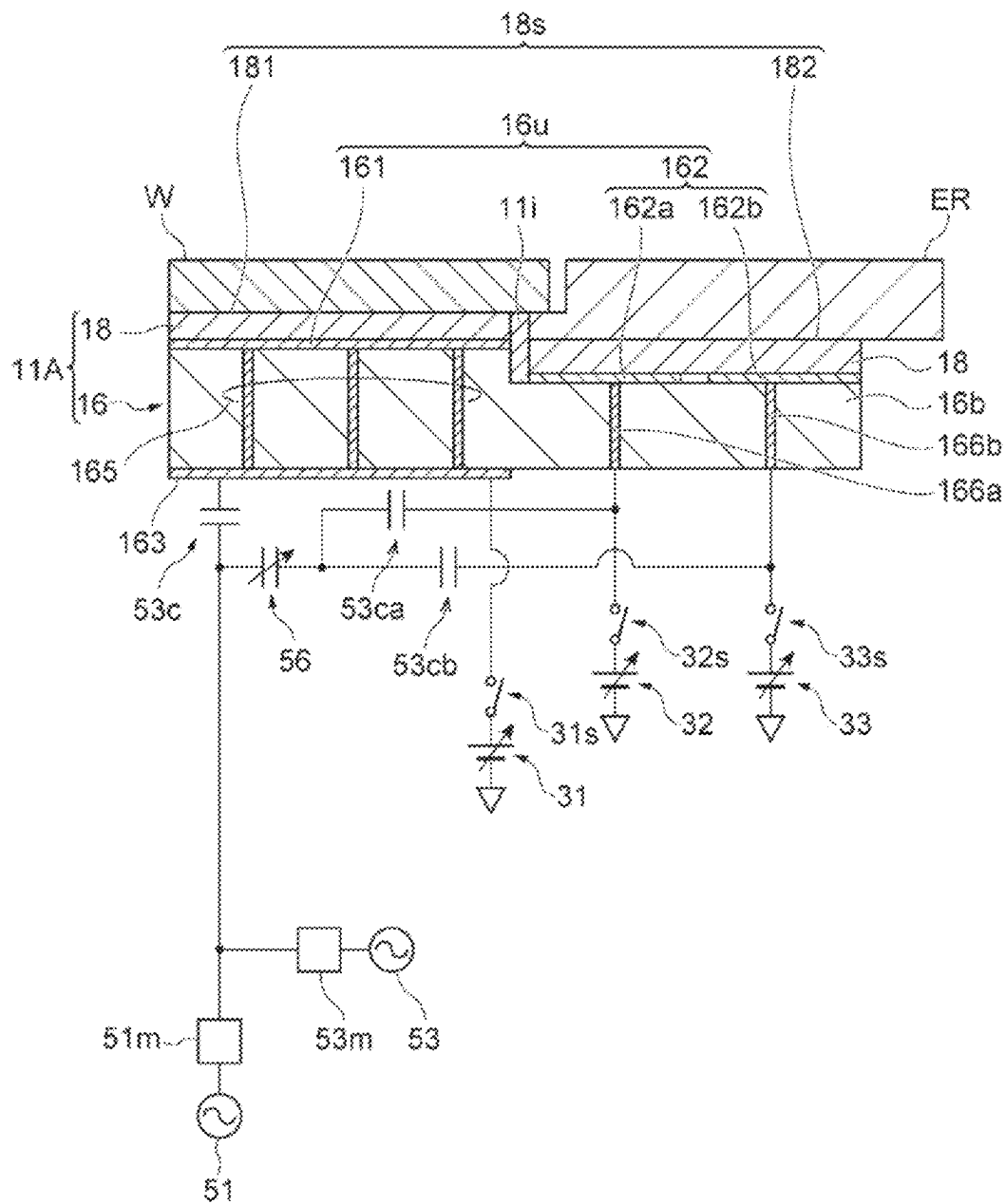


FIG. 6

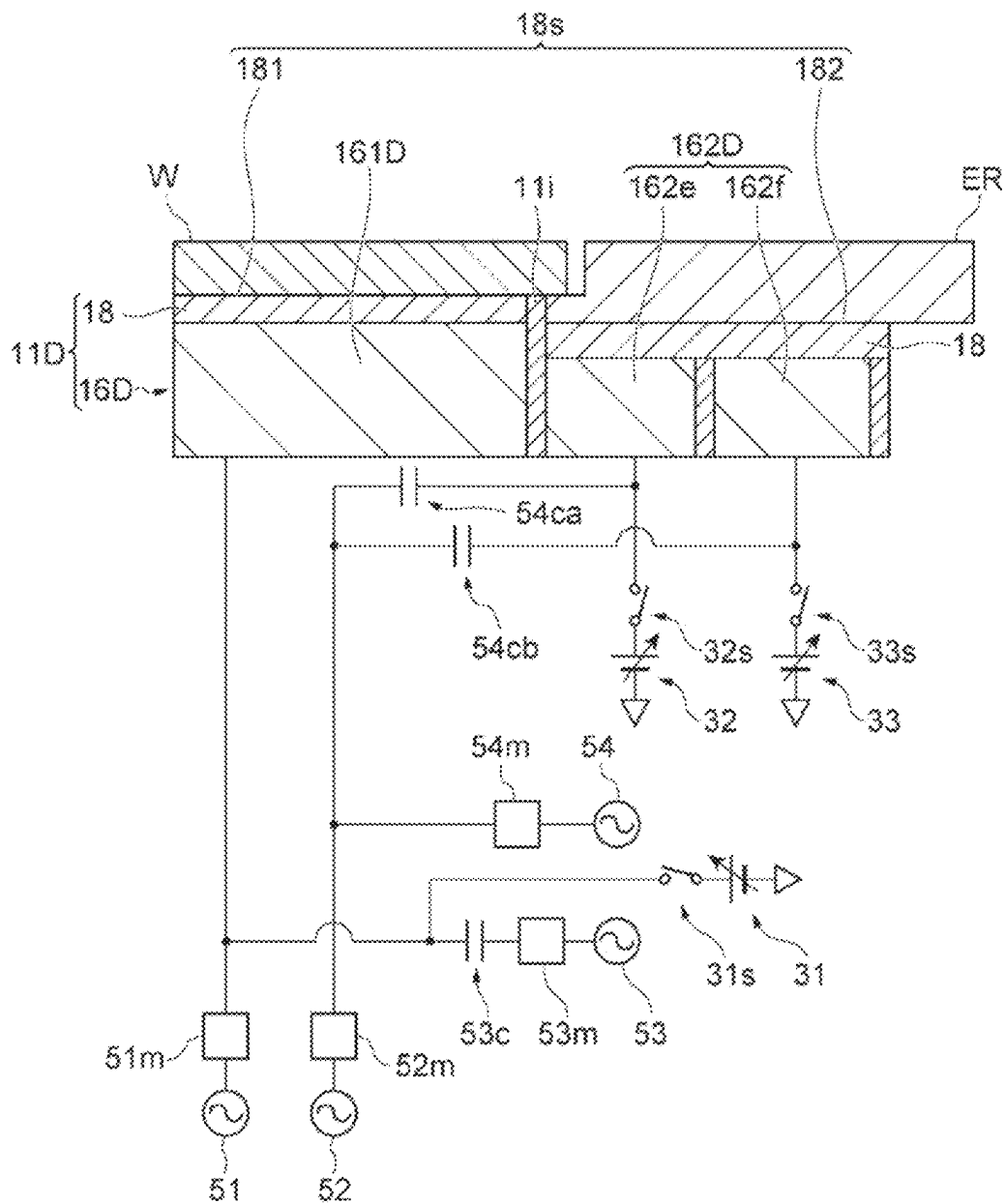


FIG. 7

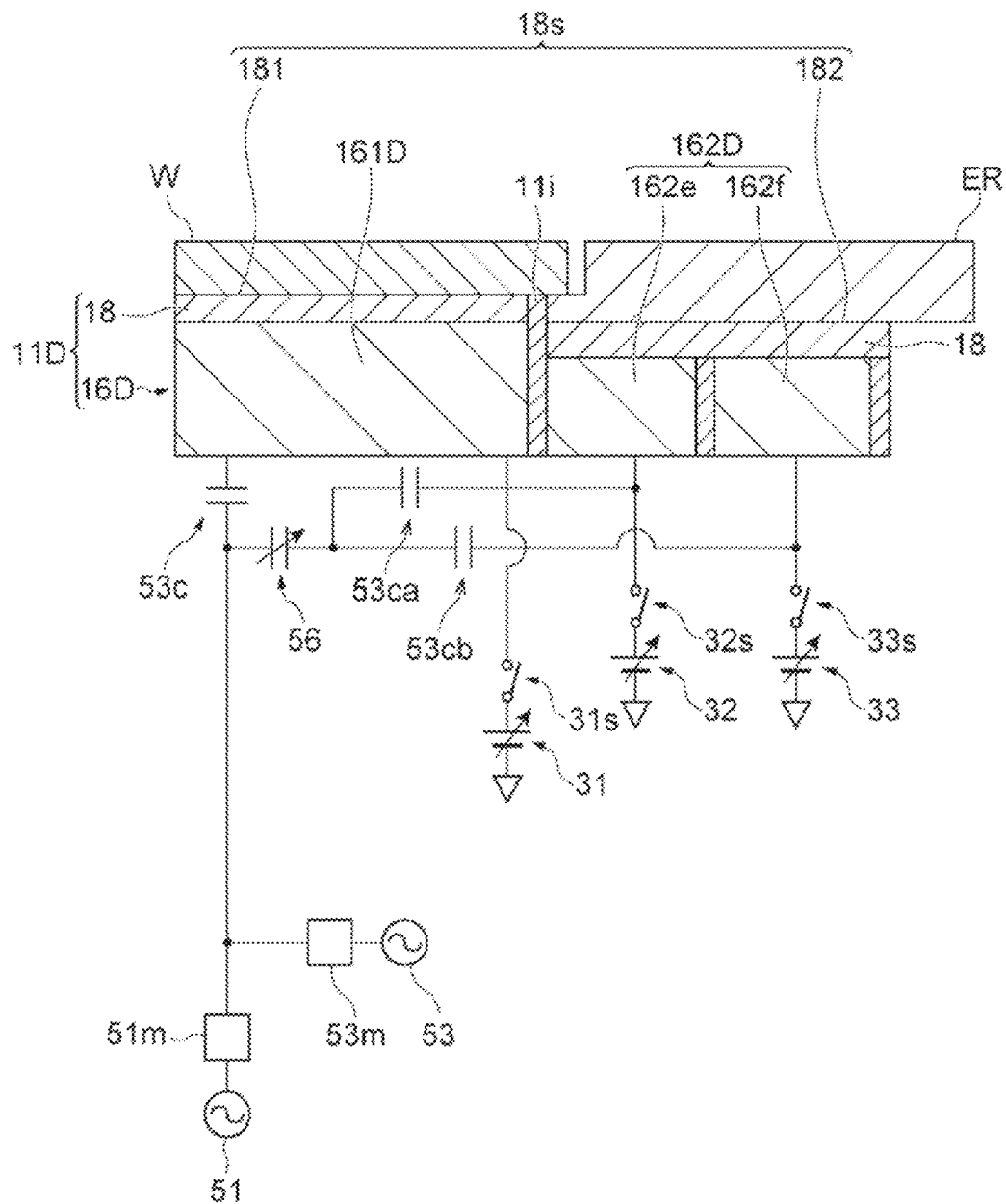


FIG. 8

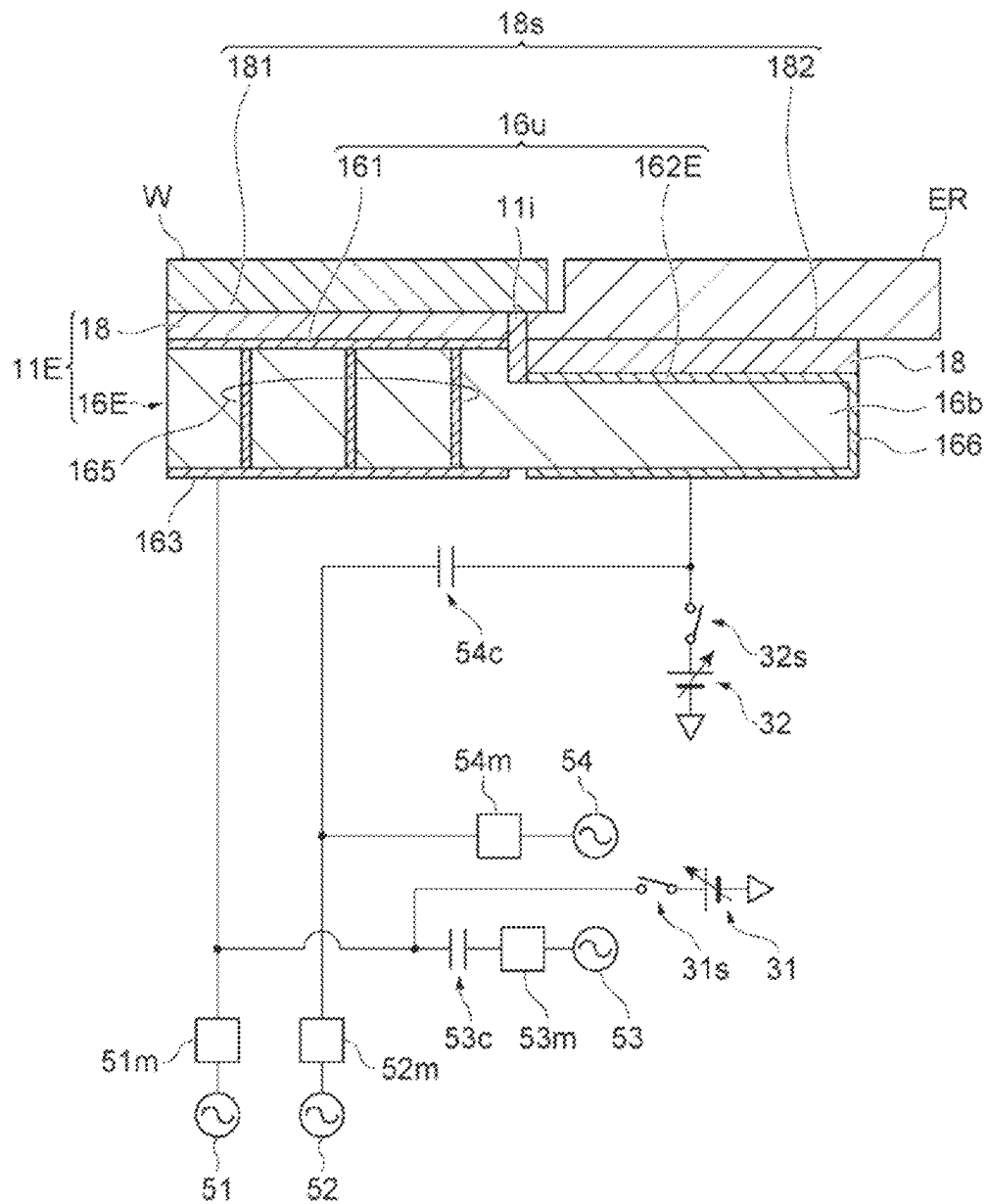


FIG. 9

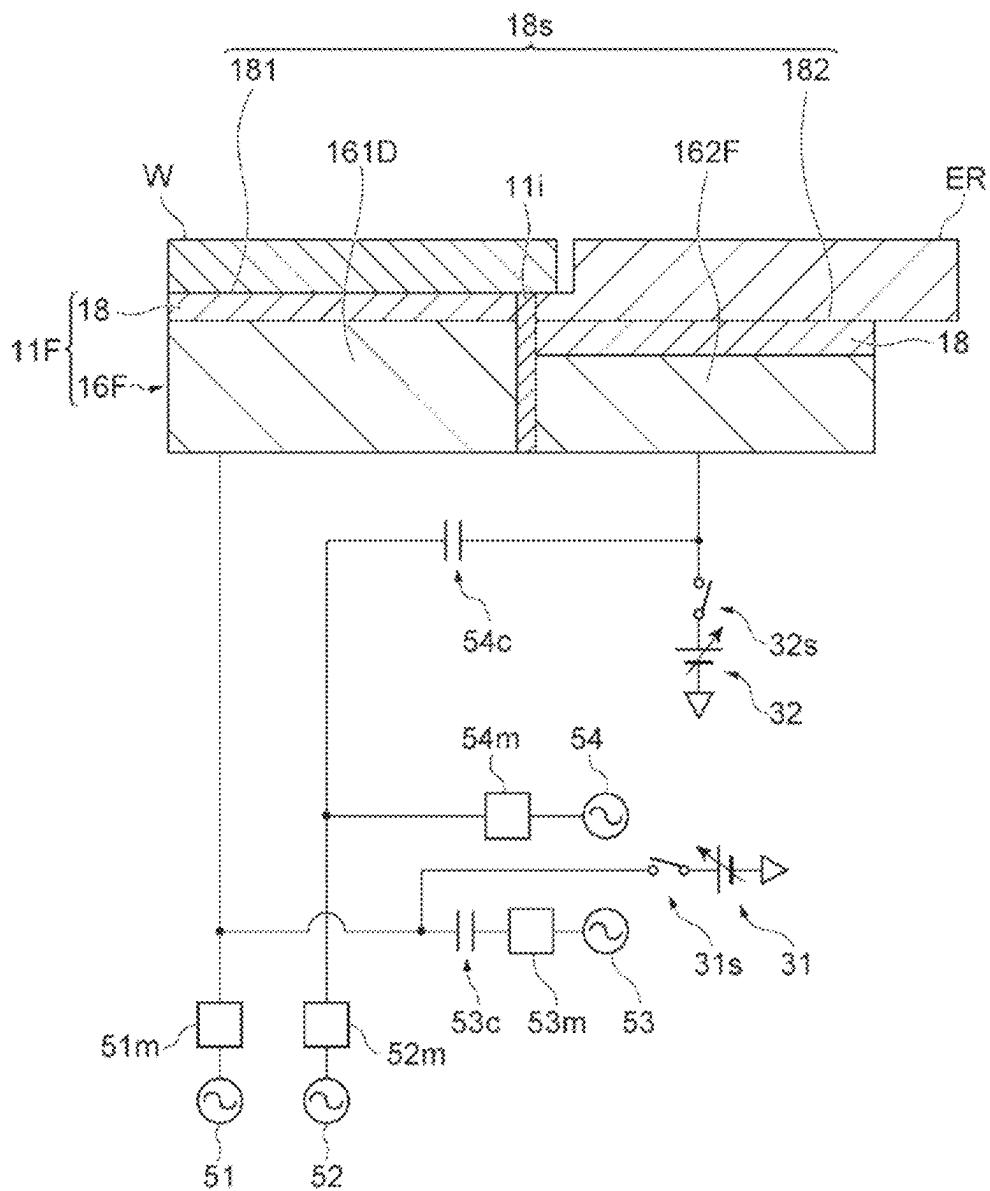


FIG. 10

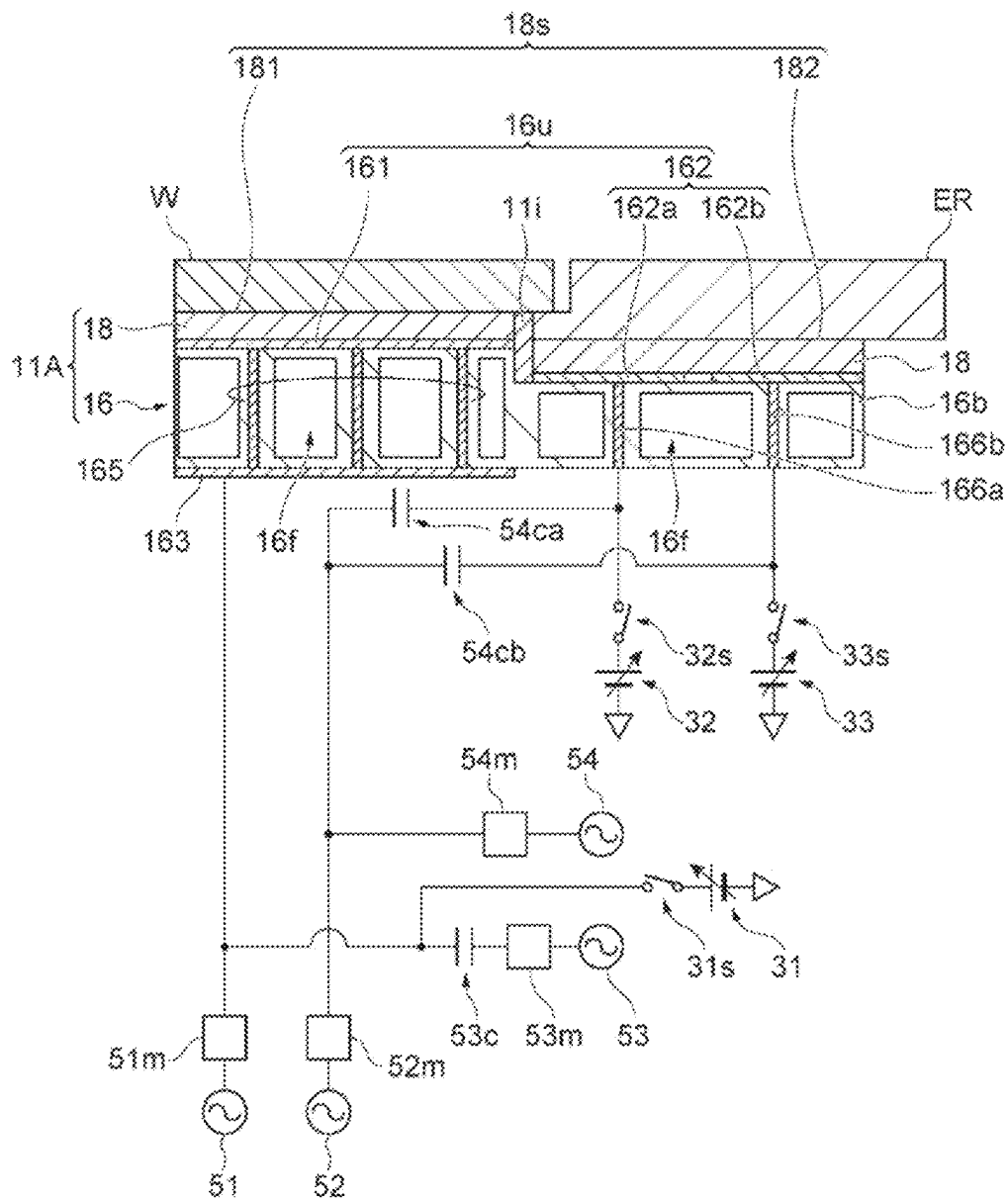


FIG. 11

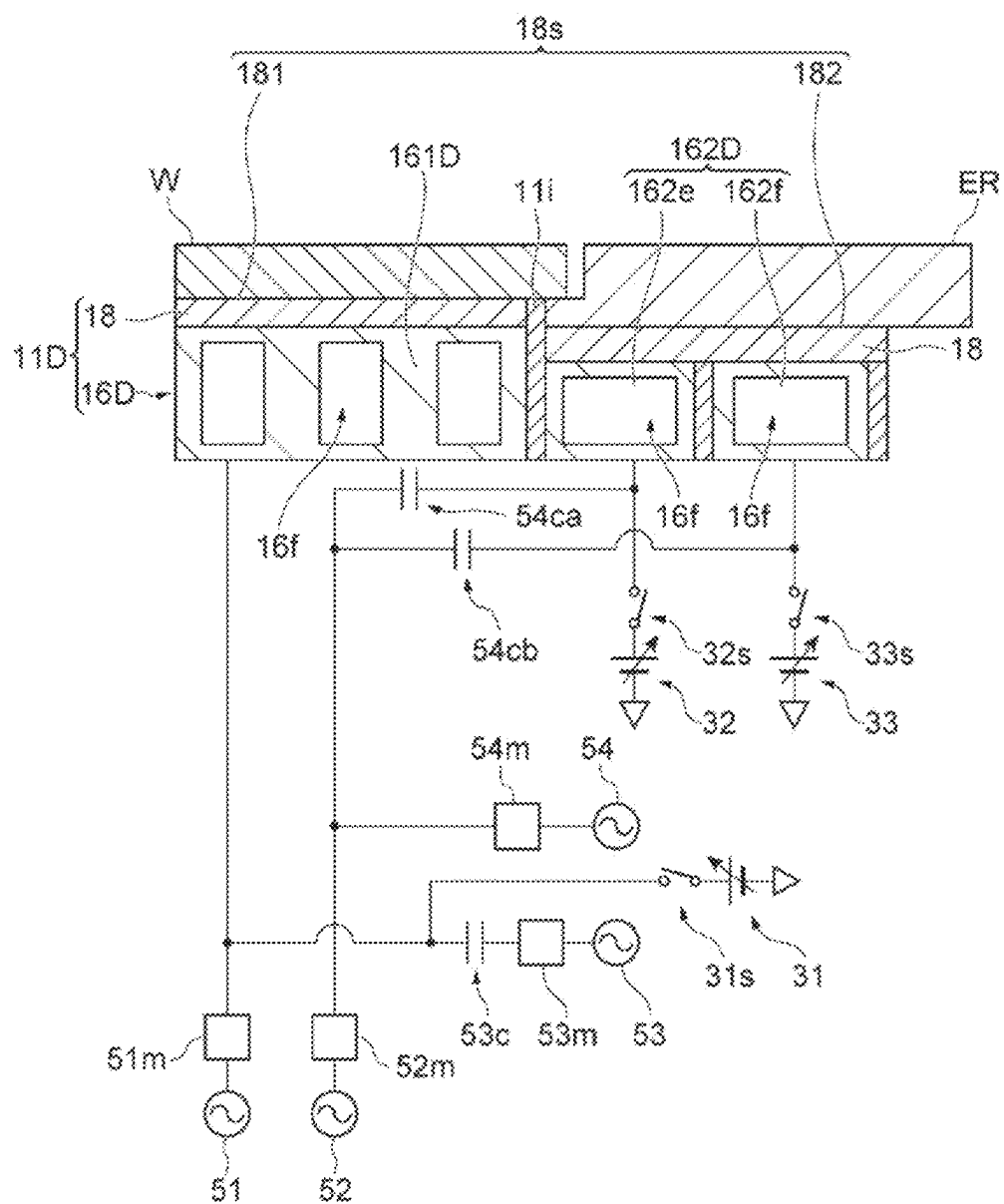
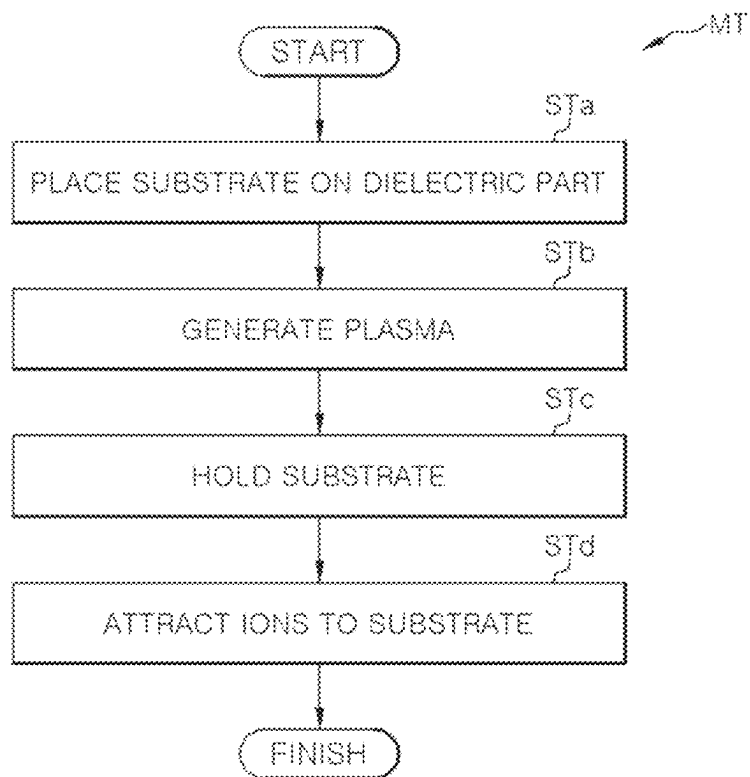


FIG. 12

1

PLASMA PROCESSING APPARATUS AND SUBSTRATE PROCESSING METHOD

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a bypass continuation application of International Application No. PCT/JP2022/016485 having an international filing date of Mar. 31, 2022 and designating the United States, the International Application being based upon and claiming the benefit of priority from Japanese Patent Application No. 2021-073411, filed on Apr. 23, 2021, the entire contents of which are incorporated herein by reference.

TECHNICAL FIELD

Exemplary embodiments of the present disclosure relate to a plasma processing apparatus and a substrate processing method.

BACKGROUND

A plasma processing apparatus is used for plasma processing of a substrate. The plasma processing apparatus includes a chamber and a substrate support. The substrate support includes a base and an electrostatic chuck. The base constitutes a lower electrode. A bias power supply is connected to the base. The electrostatic chuck is disposed on the base. The electrostatic chuck includes an insulating layer and an electrode embedded in the insulating layer. A DC power supply is connected to the electrode of the electrostatic chuck. Japanese Laid-open Patent Publication No. 2019-169635 and Japanese Laid-open Patent Publication No. 2020-205444 disclose the above-described plasma processing apparatus.

SUMMARY

The present disclosure provides a technique for efficiently supplying bias energy from a bias power supply to a substrate.

In one exemplary embodiment of the present disclosure, a plasma processing apparatus is provided. The plasma processing apparatus comprises a chamber, a substrate support, a plasma generator, a bias power supply and a chuck power supply. The substrate support is disposed in the chamber. The plasma generator is configured to generate plasma in the chamber. The bias power supply is configured to generate bias energy for attracting ions from the plasma to a substrate. The chuck power supply is configured to generate a voltage applied to the substrate support to hold the substrate using an electrostatic attractive force. The bias energy is a radio frequency power or a periodically generated voltage pulse. The substrate support includes a base and a dielectric part. The base includes a base member and an electrode. The base member is made of a dielectric or an insulator. The electrode is formed on an upper surface of the base member. The electrode forms an upper surface of the base. The dielectric part provides a supporting surface on which the substrate is placed. The dielectric part extends from the upper surface of the base to the supporting surface and is made of only a dielectric. The bias power supply and the chuck power supply are electrically connected to the electrode of the base.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 schematically shows a plasma processing apparatus according to one exemplary embodiment.

2

FIG. 2 schematically shows a plasma processing apparatus according to one exemplary embodiment.

FIG. 3 shows a substrate support and a plurality of power supplies of a plasma processing apparatus according to one exemplary embodiment.

FIG. 4 shows a substrate support and a plurality of power supplies of a plasma processing apparatus according to another exemplary embodiment.

FIG. 5 shows a substrate support and a plurality of power supplies of a plasma processing apparatus according to still another exemplary embodiment.

FIG. 6 shows a substrate support and a plurality of power supplies of a plasma processing apparatus according to further still another exemplary embodiment.

FIG. 7 shows a substrate support and a plurality of power supplies of a plasma processing apparatus according to further still another exemplary embodiment.

FIG. 8 shows a substrate support and a plurality of power supplies of a plasma processing apparatus according to further still another exemplary embodiment.

FIG. 9 shows a substrate support and a plurality of power supplies of a plasma processing apparatus according to further still another exemplary embodiment.

FIG. 10 shows a substrate support and a plurality of power supplies of a plasma processing apparatus according to further still another exemplary embodiment.

FIG. 11 shows a substrate support and a plurality of power supplies of a plasma processing apparatus according to further still another exemplary embodiment.

FIG. 12 is a flowchart of a substrate processing method according to one exemplary embodiment.

DETAILED DESCRIPTION

Hereinafter, various exemplary embodiments will be described.

In one exemplary embodiment of the present disclosure, a plasma processing apparatus is provided. The plasma processing apparatus comprises a chamber, a substrate support, a plasma generator, a bias power supply and a chuck power supply. The substrate support is disposed in the chamber. The plasma generator is configured to generate plasma in the chamber. The bias power supply is configured to generate bias energy for attracting ions from the plasma to a substrate. The chuck power supply is configured to generate a voltage applied to the substrate support to hold the substrate using an electrostatic attractive force. The bias energy is a radio frequency power or a periodically generated voltage pulse. The substrate support includes a base and a dielectric part. The base includes a base member and an electrode. The base member is made of a dielectric or an insulator. The electrode is formed on an upper surface of the base member. The electrode forms an upper surface of the base. The dielectric part provides a supporting surface on which the substrate is placed. The dielectric part extends from the upper surface of the base to the supporting surface and is made of only a dielectric. The bias power supply and the chuck power supply are electrically connected to the electrode of the base.

In the above embodiment, the electrode is not disposed in the dielectric part, so that the thickness of the dielectric part can be reduced. Therefore, the impedance between the electrode of the base and the substrate is reduced. Accordingly, the bias energy is efficiently supplied to the substrate.

In one exemplary embodiment of the present disclosure, the bias power supply may be electrically connected to the

3

electrode via a capacitor. The bias power is separated from the chuck power supply in a DC manner.

In one exemplary embodiment of the present disclosure, the bias power supply and the chuck power supply may be electrically connected to the electrode via a wiring extending between the upper surface and a bottom surface of the base member.

In one exemplary embodiment of the present disclosure, the plasma generator may include a radio frequency power supply electrically connected to the electrode of the base.

In one exemplary embodiment of the present disclosure, the supporting surface may include a first region on which the substrate is placed and a second region on which an edge ring is placed. In this embodiment, the electrode is a first electrode disposed below the first region. The base may further include a second electrode. The second electrode is disposed below the second region, forms the upper surface of the base, and is separated from the first electrode. The bias power supply or another bias power supply and another chuck power supply are electrically connected to the second electrode.

In one exemplary embodiment of the present disclosure, the second electrode may include two electrodes constituting a bipolar electrode. Said another chuck power supply may include two power supplies respectively connected to the two electrodes.

In one exemplary embodiment of the present disclosure, the bias power supply or said another bias power supply may be electrically connected to the second electrode via a capacitor. The bias power supply or said another bias power supply is separated from said another chuck power supply in a DC manner.

In one exemplary embodiment of the present disclosure, the bias power supply or said another bias power supply and said another chuck power supply may be electrically connected to the second electrode via a wiring extending between the upper surface and a bottom surface of the base member.

In one exemplary embodiment of the present disclosure, a plasma processing apparatus is provided. The plasma processing apparatus includes a chamber, a substrate support, a plasma generator, a bias power supply and a chuck power supply. The substrate support is disposed in the chamber. The plasma generator is configured to generate plasma in the chamber. The bias power supply is configured to generate bias energy for attracting ions from the plasma to a substrate. The chuck power supply is configured to generate a voltage applied to the substrate support to hold the substrate using an electrostatic attractive force. The bias energy is a radio frequency power or a periodically generated voltage pulse. The substrate support includes a base and a dielectric part. The dielectric part provides a supporting surface on which a substrate is placed, extends from an upper surface of the base to the supporting surface, and is made of only a dielectric. The supporting surface includes a first region on which the substrate is placed and a second region on which an edge ring is placed. The base includes a first portion and a second portion. The first portion is made of a conductive material and disposed below the first region. The second portion is made of a conductive material, disposed below the second region, and separated from the first portion. The bias power supply and the chuck power supply are electrically connected to the first portion. The bias power supply or another bias power supply and another chuck power supply are electrically connected to the second portion.

4

In the above embodiment, the electrode is not disposed in the dielectric part, so that the thickness of the dielectric part can be reduced. Therefore, the impedance between the first portion of the base and the substrate and the impedance between the second portion of the base and the edge ring are reduced. Accordingly, the bias energy is efficiently delivered to the substrate and the edge ring.

In one exemplary embodiment of the present disclosure, the second portion may include two portions constituting a bipolar electrode. Said another chuck power supply may include two power supplies respectively connected to the two portions.

In one exemplary embodiment of the present disclosure, the bias power supply may be electrically connected to the first portion via a capacitor. The bias power supply is separated from the chuck power supply in a DC manner. The bias power supply or said another bias power supply may be electrically connected to the second portion via a capacitor. The bias power supply or said another bias power supply is separated from the separate chuck power supply in a DC manner.

In one exemplary embodiment of the present disclosure, the plasma generator may include a radio frequency power supply electrically connected to the first portion of the base.

In another exemplary embodiment of the present disclosure, a substrate processing method using the plasma processing apparatus according to one of the exemplary embodiments is provided. The substrate processing method comprises placing a substrate on the dielectric part of the substrate support. The substrate processing method comprises generating plasma in the chamber using the plasma generator. The substrate processing method comprises holding the substrate on the substrate support by supplying a voltage from the chuck power supply. The substrate processing method comprises attracting ions from the plasma to the substrate by supplying the bias energy from the bias power supply.

Hereinafter, various exemplary embodiments will be described in detail with reference to the accompanying drawings. Further, like reference numerals will be used for like or corresponding parts throughout the drawings.

FIGS. 1 and 2 schematically show a plasma processing apparatus according to one exemplary embodiment.

In one embodiment, a plasma processing system includes a plasma processing apparatus 1 and a controller 2. The plasma processing apparatus 1 includes a plasma processing chamber 10, a substrate support 11, and a plasma generator 12. The plasma processing chamber 10 has a plasma processing space. Further, the plasma processing chamber 10 includes at least one gas inlet for supplying at least one processing gas to the plasma processing space, and at least one gas outlet for exhausting a gas from the plasma processing space. The gas inlet is connected to a gas supply part 20 to be described later, and the gas outlet is connected to an exhaust system 40 to be described later. The substrate support 11 is disposed in the plasma processing space, and has a substrate supporting surface for supporting the substrate.

The plasma generator 12 is configured to generate plasma from at least one processing gas supplied into the plasma processing space. The plasma generated in the plasma processing space includes capacitively coupled plasma (CCP), inductively coupled plasma (ICP), electron-cyclotron-resonance (ECR) plasma, and helicon wave excited (HWP) plasma, surface wave plasma (SWP), or the like. Further, various types of plasma generators including an alternating current (AC) plasma generator and a direct

5

current (DC) plasma generator may be used. In one embodiment, an AC signal (AC power) used in the AC plasma generator has a frequency within a range of 100 kHz to 10 GHz. Therefore, the AC signal includes a radio frequency (RF) signal and a microwave signal. In one embodiment, the RF signal has a frequency within a range of 200 kHz to 150 MHz.

The controller 2 processes computer-executable instructions that cause the plasma processing apparatus 1 to perform various steps described in the present disclosure. The controller 2 may be configured to control individual components of the plasma processing apparatus 1 to perform various steps described herein. In one embodiment, the controller 2 may be partially or entirely included in the plasma processing apparatus 1. The controller 2 may include, for example, a computer 2a. The computer 2a may include, for example, a central processing unit (CPU) 2a1, a storage device 2a2, and a communication interface 2a3. The central processing unit 2a1 may be configured to perform various control operations based on programs stored in the storage device 2a2. The storage device 2a2 may include a random access memory (RAM), a read only memory (ROM), a hard disk drive (HDD), a solid state drive (SSD), or a combination thereof. The communication interface 2a3 may communicate with the plasma processing apparatus 1 through a communication line such as a local area network (LAN).

Hereinafter, a configuration example of a capacitively coupled plasma processing apparatus as an example of the plasma processing apparatus 1 will be described. The capacitively coupled plasma processing apparatus 1 includes a plasma processing chamber 10, a gas supply part 20, a plurality of power supplies, and an exhaust system 40. The plasma processing apparatus 1 further includes a substrate support 11 and a gas introducing part. The gas introducing part is configured to introduce at least one processing gas into the plasma processing chamber 10. The gas introducing part includes a shower head 13. The substrate support 11 is disposed in the plasma processing chamber 10. The shower head 13 is disposed above the substrate support 11. In one embodiment, the shower head 13 forms at least a part of the ceiling of the plasma processing chamber 10. The plasma processing chamber 10 has a plasma processing space 10s defined by the shower head 13, a sidewall 10a of the plasma processing chamber 10, and the substrate support 11. The sidewall 10a is grounded. The shower head 13 and the substrate support 11 are electrically insulated from the housing of the plasma processing chamber 10.

The substrate support 11 is configured to support the substrate W placed thereon. The substrate support 11 is configured to further support an edge ring ER placed thereon. The edge ring ER is an annular member, and is made of a material such as silicon or silicon carbide. The substrate W is placed on the substrate support 11 and is disposed in the region surrounded by the edge ring ER. The substrate support 11 may include a temperature control module configured to adjust at least one of the edge ring ER and the substrate W to a target temperature. The temperature control module may include a heater, heat transfer medium, a channel, or combinations thereof. A heat transfer fluid, such as brine or a gas, flows through the channel. Further, the substrate support 11 may include a heat transfer gas supply part configured to supply a heat transfer gas to a gap between the backside of the substrate W and a substrate supporting surface 111a.

The shower head 13 is configured to introduce at least one processing gas from the gas supply part 20 into the plasma

6

processing space 10s. The shower head 13 has at least one gas supply port 13a, at least one gas diffusion space 13b, and a plurality of gas inlet ports 13c. The processing gas supplied to the gas supply port 13a passes through the gas diffusion space 13b and is introduced into the plasma processing space 10s from the gas inlet ports 13c. The shower head 13 also includes a conductive member. The conductive member of the shower head 13 functions as an upper electrode. The gas introducing part may include, in addition to the shower head 13, one or more side gas injectors (SGI) attached to one or more openings formed in the sidewall 10a.

The gas supply part 20 may include at least one gas source 21 and at least one flow rate controller 22. In one embodiment, the gas supply part 20 is configured to supply at least one processing gas from the corresponding gas source 21 to the shower head 13 through the corresponding flow rate controller 22. The flow rate controllers 22 may include, for example, a mass flow controller or a pressure-controlled flow rate controller. Further, the gas supply part 20 may include at least one flow modulation device for modulating the flow rate of at least one processing gas or causing it to pulsate.

The power supplies include one or more radio frequency power supplies constituting the plasma generator, one or more bias power supplies for generating bias energy for attracting ions to the substrate, and one or more chuck power supplies for maintaining the substrate W and the edge ring ER using an electrostatic attractive force. These power supplies will be described later.

The exhaust system 40 may be connected to a gas exhaust port 10e disposed at the bottom portion of the plasma processing chamber 10, for example. The exhaust system 40 may include a pressure control valve and a vacuum pump. The pressure in the plasma processing space 10s is adjusted by the pressure control valve. The vacuum pump may include a turbo molecular pump, a dry pump, or a combination thereof.

Hereinafter, FIG. 3 will be referred to. FIG. 3 shows a substrate support and a plurality of power supplies of a plasma processing apparatus according to one exemplary embodiment. The substrate support section 11A shown in FIG. 3 may be employed as the substrate support section 11 of the plasma processing apparatus 1.

A substrate support 11A includes a base 16 and a dielectric part 18. The base 16 includes a base member 16b and one or more electrodes. The base member 16b is made of a dielectric or an insulator. The base member 16b has a substantially disc shape. The base member 16b is made of ceramic, such as aluminum nitride or aluminum oxide.

One or more electrodes of the base 16 are formed on the upper surface of the base member 16b. One or more electrodes of the base 16 constitute an upper surface 16u of the base 16. One or more electrodes of base 16 may be conductive films. In one embodiment, the base 16 may include, as one or more electrodes, a first electrode 161 and a second electrode 162. The first electrode 161 includes the center of the upper surface 16u, and has a substantially circular shape. The second electrode 162 is separated from the first electrode 161. The second electrode 162 may extend in a circumferential direction to surround the first electrode 161. In other words, the second electrode 162 may have an annular shape.

In one embodiment, the second electrode 162 may include two electrodes 162a and 162b constituting a bipolar electrode. The two electrodes 162a and 162b are separated from each other. Each of the two electrodes 162a and 162b may have an annular shape. One of the two electrodes 162a and

162b may extend in the circumferentially direction at a radially outer side of the other electrode. Alternatively, each of the two electrodes **162a** and **162b** may include a plurality of electrodes that are separated from each other and arranged in the circumferential direction.

In one embodiment, the base **16** may include an electrode **163**. The electrode **163** may be a conductive film formed on the bottom surface of the base member **16b**. In one embodiment, the base **16** may include a plurality of wirings **165**, **166a**, and **166b**. The wirings **165**, **166a**, and **166b** extend between the upper surface and the bottom surface of the base member **16b**. One or more wirings **165** connect the first electrode **161** and the electrode **163**. One or more wirings **166a** are connected to the electrode **162a**. One or more wirings **166b** are connected to the electrode **162b**. Each of the wirings **165**, **166a**, and **166b** may be a via hole formed in the base member **16b**. Alternatively, each of the wirings **165**, **166a**, and **166b** may be a conductor line formed along the surface of the base member **16b**. The electrode **163** may be a planar electrode, or may be a wiring that connects the power supply points for the first electrode **161**, the electrode **162a**, and the electrode **162b** to the wiring **165**, **166a**, and **166b**, respectively.

The dielectric part **18** provides a supporting surface **18s** on which the substrate **W** is placed. The dielectric part **18** extends from the upper surface **16u** of the base **16** to the supporting surface **18s**, and is made of only a dielectric. The dielectric part **18** is made of ceramic such as aluminum nitride or aluminum oxide.

In one embodiment, the supporting surface **18s** of the dielectric part **18** may include a first region **181** and a second region **182**. The first region **181** includes the center of the supporting surface **18s**, and has a substantially circular shape. The substrate **W** is placed on the first region **181**. The first region **181** extends above the first electrode **161** of the base **16**. In other words, the first electrode **161** is disposed below the first region **181**.

The second region **182** extends in the circumferential direction at a radially outer side of the first region **181**. In other words, the second region **182** has an annular shape. The edge ring **ER** is placed on the second region **182**. The second region **182** extends above the second electrode **162** of the base **16**. In other words, the second electrode **162** (the electrodes **162a** and **162b**) is disposed below the second region **182**.

In one embodiment, the substrate support **11** may further include an insulator part **11i**. The insulator part **11i** covers the end surface (the outer circumferential surface) of the first electrode **161** and the inner end surface (the inner circumferential surface) of the second electrode **162** to hide them from plasma. In one embodiment, the dielectric part **18** may be separated into a portion providing the first region **181** and a portion providing the second region **182**. The insulator part **11i** may be interposed between the outer circumferential surface of the portion providing the first region **181** and the inner circumferential surface of the portion providing the second region **182**.

The power supplies of the plasma processing apparatus **1** may include chuck power supplies **31**, **32**, and **33**. Each of the chuck power supplies **31**, **32**, and **33** is a DC power supply or a variable DC power supply. The chuck power supply **31** is connected to the first electrode **161** via a switch **31s**, an electrode **163**, and one or more wirings **165**. When a DC voltage from the chuck power supply **31** is applied to the first electrode **161**, the substrate **W** is attracted to the first region **181** by the electrostatic attractive force and is held on the substrate support **11A**.

The chuck power supply **32** is connected to the electrode **162a** via a switch **32s** and one or more wirings **166a**. The chuck power supply **33** is connected to the electrode **162b** via a switch **33s** and one or more wirings **166b**. When a DC voltage from the chuck power supply **32** is applied to the electrode **162a** and a DC voltage from the chuck power supply **33** is applied to the electrode **162b**, the edge ring **ER** is attracted to the second region **182** by the electrostatic attractive force and is held on the substrate support **11A**.

The power supplies of the plasma processing apparatus **1** include one or more radio frequency power supplies and one or more bias power supplies. In the example shown in FIG. **3**, the power supplies include a radio frequency power supply **51**, a radio frequency power supply **52**, a bias power supply **53**, and a bias power supply **54**.

The radio frequency power supply **51** and the radio frequency power supply **52** constitute a plasma generator of one embodiment. Each of the radio frequency power supply **51** and the radio frequency power supply **52** generates a radio frequency power having a frequency suitable for generating plasma from a gas in the chamber **10**. The radio frequency power generated by each of the radio frequency power supplies **51** and **52** has a frequency within a range of 13 MHz to 150 MHz, for example.

The radio frequency power supply **51** is connected to the first electrode **161** via a matching device **51m**, an electrode **163**, and one or more wirings **165**. The matching device **51m** includes a matching circuit for matching an impedance of a load of the radio frequency power supply **51** with an output impedance of the radio frequency power supply **51**.

The radio frequency power supply **52** is connected to the electrode **162a** via a matching device **52m**, a capacitor **54ca**, and one or more wirings **166a**. Further, the radio frequency power supply **52** is connected to the electrode **162b** via the matching device **52m**, a capacitor **54cb**, and one or more wirings **166b**. The matching device **52m** includes a matching circuit for matching the impedance of the load of the radio frequency power supply **52** with the output impedance of the radio frequency power supply **52**. The capacitor **54ca** separates the radio frequency power supply **52** from the chuck power supply **32** in a DC manner. The capacitor **54cb** separates the radio frequency power supply **52** from the chuck power supply **33** in a DC manner.

The bias power supplies **53** and **54** generate bias energy for attracting ions to the substrate **W** and the edge ring **ER**. The bias energy may be a radio frequency bias power. The radio frequency bias power has a frequency within a range of 100 kHz to 13.56 MHz, for example. Alternatively, the bias energy may be periodically generated voltage pulses. The voltage pulse has positive polarity or negative polarity. The voltage pulse may have any waveform. The voltage pulse may be a negative DC voltage pulse. The voltage pulse is generated periodically with a repetition frequency within a range of 100 kHz to 13.56 MHz.

The bias power supply **53** is connected to the first electrode **161** via a matching device **53m**, a capacitor **53c**, an electrode **163**, and one or more wirings **165**. The matching device **53m** includes a matching circuit for matching the impedance of the load of the bias power supply **53** with the output impedance of the bias power supply **53**. The capacitor **53c** is provided to separate the bias power supply **53** from the chuck power supply **31** in a DC manner.

The bias power supply **54** is connected to the electrode **162a** via a matching device **54m**, a capacitor **54ca**, and one or more wirings **166a**. Further, the bias power supply **54** is connected to the electrode **162b** via a matching device **54m**, a capacitor **54cb**, and one or more wirings **166b**. The

matching device **54m** includes a matching circuit for matching the impedance of the load of the bias power supply **54** with the output impedance of the bias power supply **54**. The capacitor **54ca** is provided to separate the bias power supply **54** from the chuck power supply **32** in a DC manner. The capacitor **54cb** is provided to separate the bias power supply **54** from the chuck power supply **33** in a DC manner.

In the plasma processing apparatus **1**, no electrode is disposed in the dielectric part **18**, so that the thickness of the dielectric part **18** can be reduced. Therefore, the impedance between the first electrode **161** of the base **16** and the substrate **W** is reduced. Further, the impedance between each of the electrodes **162a** and **162b** of the base **16** and the edge ring **ER** is reduced. Accordingly, the bias energy is efficiently supplied to the substrate **W**. Further, the bias energy is efficiently supplied to the edge ring **ER**. Moreover, the radio frequency power is efficiently coupled to the plasma via the substrate **W** and the edge ring **ER**. Thus, it is possible to reduce the bias energy. Further, it is possible to reduce the radio frequency power. Hence, heat generation at the electrodes of the substrate support **11A**, the contact points, and the like is suppressed.

Since the base member **16b** of the base **16** is made of a dielectric or an insulator, the difference between the thermal expansion coefficient of the base **16** and the thermal expansion coefficient of the dielectric part **18** is small. Therefore, the damage to the substrate support **11A** due to the difference between the thermal expansion coefficient of the base **16** and the thermal expansion coefficient of the dielectric part **18** is suppressed.

In one embodiment, the dielectric part **18** may satisfy the following Eq. (1).

$$0.5 \times C_{W0}/S_W < C_{FO}/S_F < 1.5 \times C_{W0}/S_W \quad \text{Eq. (1)}$$

In Eq. (1), C_{W0} indicates the capacitance between the base **16** and the substrate **W**. S_W indicates the area of one main surface (for example, the bottom surface or the surface in contact with plasma) of the substrate **W**. C_{FO} indicates the capacitance between the base **16** and the edge ring **ER**. S_F indicates the area of one main surface (for example, the bottom surface or the surface in contact with plasma) of the edge ring. By satisfying Eq. (1), the difference between the power density of the radio frequency power coupled to the plasma via the substrate **W** and the power density of the radio frequency power coupled to the plasma via the edge ring **ER** is reduced. Therefore, the variation in the plasma density in the chamber is suppressed.

In one embodiment, the substrate support **11A** may be configured such that the capacitance between the substrate **W** and the edge ring **ER** becomes 10 nF or less or 3 nF or less. Accordingly, the electrical coupling between the substrate **W** and the edge ring **ER** is suppressed.

Hereinafter, FIG. 4 will be referred to. FIG. 4 shows a substrate support and a plurality of power supplies of a plasma processing apparatus according to another exemplary embodiment. A substrate support **11B** shown in FIG. 4 may be employed as the substrate support **11** of the plasma processing apparatus **1**. The substrate support **11B** further includes one or more wirings **165a** and one or more wirings **165b**. One or more wirings **165a** and one or more wirings **165b** extend between the upper surface and the bottom surfaces of the base member **16b**. One or more wirings **165a** connect the electrode **163** and the electrode **162a**. One or more wiring **165b** connects the electrode **163** and the electrode **162b**.

Each of one or more wirings **165a** and one or more wirings **165b** may be a via hole formed in the base member

16b. Alternatively, each of one or more wirings **165a** and one or more wirings **165b** may be a conductor line formed along the surface of the base member **16b**.

In the example shown in FIG. 4, the plasma processing apparatus **1** does not include the radio frequency power supply **52** and the matching device **52m**. In the example shown in FIG. 4, the radio frequency power supply **51** is connected to the electrode **162a** via the matching device **51m**, the electrode **163**, and one or more wirings **165a**. Further, the radio frequency power supply **51** is connected to the electrode **162b** via the matching device **51m**, the electrode **163**, and one or more wirings **165b**. In the example shown in FIG. 4, the radio frequency power from the single radio frequency power supply **51** is distributed to the first electrode **161** and the second electrode **162**. The other configurations in the example shown in FIG. 4 are the same as those of the example shown in FIG. 3.

Hereinafter, FIG. 5 will be referred to. FIG. 5 shows a substrate support and a plurality of power supplies of a plasma processing apparatus according to still another exemplary embodiment. As shown in FIG. 5, the plasma processing apparatus **1** may not include the radio frequency power supply **52**, the matching device **52m**, the bias power supply **54**, and the matching device **54m**.

As shown in FIG. 5, the radio frequency power supply **51** is connected to the electrode **162a** via the matching device **51m**, an impedance circuit **56**, a capacitor **53ca**, and the wiring **166a**. Further, the radio frequency power supply **51** is connected to the electrode **162b** via the matching device **51m**, the impedance circuit **56**, a capacitor **53cb**, and the wiring **166b**. The bias power supply **53** is connected to the electrode **162a** via the matching device **53m**, the impedance circuit **56**, the capacitor **53ca**, and the wiring **166a**. Further, the bias power supply **53** is connected to the electrode **162b** via the matching device **53m**, the impedance circuit **56**, the capacitor **53cb**, and the wiring **166b**. The capacitor **53ca** is provided to separate the radio frequency power supply **51** and the bias power supply **53** from the chuck power supply **32** in a DC manner. The capacitor **53cb** is provided to separate the radio frequency power supply **51** and the bias power supply **53** from the chuck power supply **33** in a DC manner.

The impedance circuit **56** has a variable impedance. The impedance circuit includes a variable impedance element, such as a variable capacitor. In the plasma processing apparatus **1** of the example shown in FIG. 5, the radio frequency power from the single radio frequency power supply **51** is distributed to the first electrode **161** and the second electrode **162**. Further, in the plasma processing apparatus **1** of the example shown in FIG. 5, the bias energy from the single bias power supply **53** is distributed to the first electrode **161** and the second electrode **162**. A distribution ratio of each of the radio frequency power and the bias energy is set by adjusting the variable impedance of the impedance circuit **56**. The other configurations in the example shown in FIG. 5 are the same as those in the example shown in FIG. 3.

Hereinafter, FIG. 6 will be referred to. FIG. 6 shows a substrate support and a plurality of power supplies of a plasma processing apparatus according to further still another exemplary embodiment. A substrate support **11D** shown in FIG. 6 may be employed as the substrate support **11** of the plasma processing apparatus **1**. The substrate support **11D** has a base **16D** instead of the base **16**. The base **16D** includes a first portion **161D** and a second portion **162D**.

11

Each of the first portion **161D** and the second portion **162D** are made of a conductive material. Each of the first portion **161D** and the second portion **162D** is made of a metal such as aluminum or a metal-ceramic composite.

The first portion **161D** is disposed below the first region **181**. The first portion **161D** includes the center of the base **16D** and has a substantially disc shape. The second portion **162D** is disposed below the second region **182**. The second portion **162D** extends in the circumferential direction at a radially outer side of the first portion **161D**. The second portion **162D** may have a substantially annular shape in plan view. The second portion **162D** is separated from the first portion **161D**. The gap between the first portion **161D** and the second portion **162D** may be filled with an insulator material (the insulator part **11i**) or a dielectric material.

In one embodiment, the second portion **162D** may include portions **162e** and **162f**. The portions **162e** and **162f** constitute a bipolar electrode. Each of portions **162e** and **162f** extends in the circumferential direction at the radially outer side of the first portion **161D**. Each of the portions **162e** and **162f** may have a substantially annular shape in plan view. The portion **162f** may extend in the circumferential direction at the radially outer side of the portion **162e**. The portions **162e** and **162f** are separated from each other. The gap between the portions **162e** and **162f** may be filled with an insulator material or a dielectric material.

The chuck power supply **31** is connected to the first portion **161D** via the switch **31s**. When a DC voltage from the chuck power supply **31** is applied to the first portion **161D**, the substrate **W** is attracted to the first region **181** by an electrostatic attractive force and held on the substrate support **11D**.

The chuck power supply **32** is connected to the portion **162e** via the switch **32s**. The chuck power supply **33** is connected to the portion **162f** via the switch **33s**. When a DC voltage from the chuck power supply **32** is applied to the portion **162e** and a DC voltage from the chuck power supply **33** is applied to the portion **162f**, the edge ring **ER** is attracted to the second region **182** by an electrostatic attractive force and held on the substrate support **11D**.

The radio frequency power supply **51** is connected to the first portion **161D** via the matching device **51m**. The radio frequency power supply **52** is connected to the portion **162e** via the matching device **52m** and the capacitor **54ca**. Further, the radio frequency power supply **52** is connected to the portion **162f** via the matching device **52m** and the capacitor **54cb**.

The bias power supply **53** is connected to the first portion **161D** via the matching device **53m** and the capacitor **53c**. The bias power supply **54** is connected to the portion **162e** via the matching device **54m** and the capacitor **54ca**. The bias power supply **54** is connected to the portion **162f** via the matching device **54m** and the capacitor **54cb**.

In the substrate support **11D**, no electrode is disposed in the dielectric part **18**, so that the thickness of the dielectric part **18** can be reduced. Therefore, the impedance between the first portion **161D** of the base **16D** and the substrate **W** and the impedance between the second portion **162D** of the base **16D** and the edge ring **ER** are reduced. Accordingly, the bias energy is efficiently supplied to the substrate **W** and the edge ring **ER**.

In one embodiment, the dielectric part **18** may satisfy the following Eq. (2).

$$0.5 \times C_{W0}/S_W < C_{FO}/S_F < 1.5 \times C_{W0}/S_W \quad \text{Eq. (2)}$$

In Eq. (2), C_{W0} indicates the capacitance between the base **16D** and the substrate **W**. S_W indicates the area of one main

12

surface (for example, the bottom surface or the surface in contact with plasma) of the substrate **W**. C_{FO} indicates the capacitance between the base **16D** and the edge ring **ER**. S_F indicates the area of one main surface (for example, the bottom surface or the surface in contact with plasma) of the edge ring **ER**. By satisfying Eq. (2), the difference between the power density of the radio frequency power coupled to the plasma via the substrate **W** and the power density of the radio frequency power coupled to the plasma via the edge ring **ER** is reduced. Therefore, the variation in the plasma density in the chamber is suppressed.

In one embodiment, the substrate support **11D** may be configured such that the capacitance between the substrate **W** and the edge ring **ER** becomes 10 nF or less or 3 nF or less. Accordingly, the electrical coupling between the substrate **W** and the edge ring **ER** is suppressed.

Hereinafter, FIG. 7 will be referred to. FIG. 7 shows a substrate support and a plurality of power supplies of a plasma processing apparatus according to further still another exemplary embodiment. As shown in FIG. 7, the plasma processing apparatus **1** may not include the radio frequency power supply **52**, the matching device **52m**, the bias power supply **54**, and the matching device **54m**.

As shown in FIG. 7, the radio frequency power supply **51** is connected to the portion **162e** via the matching device **51m**, the impedance circuit **56**, and the capacitor **53ca**. Further, the radio frequency power supply **51** is connected to the portion **162f** via the matching device **51m**, the impedance circuit **56**, and the capacitor **53cb**. The bias power supply **53** is connected to the portion **162e** via the matching device **53m**, impedance circuit **56**, and the capacitor **53ca**. Further, the bias power supply **53** is connected to the portion **162f** via the matching device **53m**, the impedance circuit **56**, and the capacitor **53cb**.

In the example plasma processing apparatus **1** shown in FIG. 7, the radio frequency power from the single radio frequency power supply **51** is distributed to the first portion **161D** and the second portion **162D**. Further, in the example plasma processing apparatus **1** shown in FIG. 7, the bias energy from the single bias power supply **53** is distributed to the first portion **161D** and the second portion **162D**. A distribution ratio of each of the radio frequency power and the bias energy is set by adjusting the variable impedance of the impedance circuit **56**. The other configurations in the example shown in FIG. 7 are the same as those in the example shown in FIG. 6.

Hereinafter, FIG. 8 will be referred to. FIG. 8 shows a substrate support and a plurality of power supplies of a plasma processing apparatus according to further still another exemplary embodiment. In the substrate support **11A** and the substrate support **11B** described above, the second electrode **162** includes two electrodes. On the other hand, in the substrate support **11E** shown in FIG. 8, the second electrode **162E** is configured as a single electrode. The wiring **166** extending from the bottom surface of the base member **16b** while passing through the end surface is connected to the second electrode **162E**. Each of one or more wirings **165** may connect the electrode **163** and the first electrode **161** via a groove formed at the peripheral portion of the base member **16b**.

The chuck power supply **32** is connected to the second electrode **162E** via the switch **32s** and the wiring **166**. Further, the radio frequency power supply **52** is connected to the second electrode **162E** via the matching device **52m**, a capacitor **54c**, and the wiring **166**. Further, the bias power supply **54** is connected to the second electrode **162E** via the matching device **54m**, the capacitor **54c**, and the wiring **166**.

13

The radio frequency power from the radio frequency power supply 51 may be distributed to the first electrode 161 and the second electrode 162E. Further, the bias energy from the bias power supply 53 may be distributed to the first electrode 161 and the second electrode 162E.

Hereafter, FIG. 9 will be referred to. FIG. 9 shows a substrate support and a plurality of power supplies of a plasma processing apparatus according to further still another exemplary embodiment. In the substrate support 11D described above, the second portion 162D includes the two portions 162e and 162f. On the other hand, in the base 16F of the substrate support 11F shown in FIG. 9, a second portion 162F is formed as a single portion.

The chuck power supply 32 is connected to the second portion 162F via the switch 32s. Further, the radio frequency power supply 52 is connected to the second portion 162F via the matching device 52m and the capacitor 54c. Further, the bias power supply 54 is connected to the second portion 162F via the matching device 54m and the capacitor 54c. The radio frequency power from the radio frequency power supply 51 may be distributed to the first portion 161D and the second portion 162F. Further, the bias energy from the bias power supply 53 may be distributed to the first portion 161D and the second portion 162F.

Hereinafter, FIG. 10 will be referred to. FIG. 10 shows a substrate support and a plurality of power supplies of a plasma processing apparatus according to further still another exemplary embodiment. The example shown in FIG. 10 is a modification of the example shown in FIG. 3. In the example shown in FIG. 10, a channel 16f is formed in the base 16. A heat medium (for example, a coolant) is supplied from a supply device to the channel 16f. The heat medium flows through the channel 16f and returns to the supply device. The channel 16f may be formed to have a wide cross-sectional area without decreasing the mechanical strength of the base 16. The channel 16f may be formed in the base 16 in the above-described examples other than the example shown in FIG. 3.

Hereafter, FIG. 11 will be referred to. FIG. 11 shows a substrate support and a plurality of power supplies of a plasma processing apparatus according to further still another exemplary embodiment. The example shown in FIG. 11 is a modification of the example shown in FIG. 6. In the example shown in FIG. 11, the channel 16f is formed in the base 16D. A heat medium (for example, a coolant) is supplied from a supply device to the channel 16f. The heat medium flows through the channel 16f and returns to the supply device. The channel 16f may be formed to have a wide cross-sectional area without decreasing the mechanical strength of the base 16D. The channel 16f may be formed in the base 16D in the above-mentioned examples other than the example shown in FIG. 6. The channel 16f may be formed in the base 16F in the example shown in FIG. 9.

Hereinafter, FIG. 12 will be referred to. FIG. 12 is a flowchart of a substrate processing method according to one exemplary embodiment. The substrate processing method shown in FIG. 12 (hereinafter, referred to as "method MT") can be performed using the plasma processing apparatus 1 described above. In the method MT, the individual components of the plasma processing apparatus 1 can be controlled by the controller 2. The method MT includes steps STa, STb, STc, and STd.

In step STa, the substrate W is placed on the dielectric part 18 of the substrate support. Steps STb, STc, and STd are performed in a state where the substrate W is placed on the dielectric part 18.

14

In step STb, plasma is generated in the chamber 10 using the plasma generator. In step STb, the processing gas is supplied from the gas supply part 20 into the plasma processing chamber 10. Further, in step STb, the exhaust system 40 reduces a pressure in the plasma processing chamber 10 to a specified pressure. Further, in step STb, the radio frequency power is supplied to generate plasma from the processing gas.

In step STc, the substrate W is held by the substrate support by applying a voltage from the chuck power supply 31. The edge ring ER may be held by the substrate support in step STc by applying a voltage from the chuck power supply 32 and the chuck power supply 33. Alternatively, the edge ring ER may be held by the substrate support before step STb.

Step STd is performed when the plasma is generated in step STb. In step STd, the bias energy is supplied to attract ions from the plasma to the substrate W.

While various exemplary embodiments have been described above, the present disclosure is not limited to the above-described embodiments, and various additions, omissions, substitutions and changes may be made. Further, other embodiments can be implemented by combining elements in different embodiments.

For example, the radio frequency power from the radio frequency power supply may be supplied to the upper electrode instead of the substrate support. Further, in another embodiment, the plasma processing apparatus may be a plasma processing apparatus other than a capacitively coupled plasma processing apparatus. Such a plasma processing apparatus may be an inductively coupled plasma processing apparatus, an electron cyclotron resonance (ECR) plasma processing apparatus, or a plasma processing apparatus for generating plasma using surface waves such as microwaves.

From the above description, it will be appreciated that various embodiments of the present disclosure have been described herein for purposes of illustration, and that various changes may be made without departing from the scope and spirit of the present disclosure. Therefore, the various embodiments disclosed herein are not intended to be limiting, with a true scope and spirit being indicated by the following claims.

The invention claimed is:

1. A plasma processing apparatus comprising:

- a chamber;
 - a substrate support disposed in the chamber;
 - a plasma generator configured to generate plasma in the chamber;
 - a bias power supply configured to generate bias energy for attracting ions from the plasma to a substrate; and
 - a chuck power supply configured to generate a voltage applied to the substrate support to hold the substrate using an electrostatic attractive force,
- wherein the bias energy is a radio frequency power or a periodically generated voltage pulse,
- the substrate support includes:
- a base including a base member made of a dielectric or an insulator and an electrode formed on an upper surface of the base member, the electrode forming an upper surface of the base; and
 - a dielectric part that provides a supporting surface on which the substrate is placed, extends from the upper surface of the base to the supporting surface, and is made of only a dielectric, and
- the bias power supply and the chuck power supply are electrically connected to the electrode of the base.

15

2. The plasma processing apparatus of claim 1, wherein the bias power supply is electrically connected to the electrode via a capacitor, and is separated from the chuck power supply in a DC manner.

3. The plasma processing apparatus of claim 1, wherein the bias power supply and the chuck power supply are electrically connected to the electrode via a wiring extending between the upper surface and a bottom surface of the base member.

4. The plasma processing apparatus of claim 1, wherein the plasma generator includes a radio frequency power supply electrically connected to the electrode of the base.

5. The plasma processing apparatus of claim 1, wherein the supporting surface includes a first region on which the substrate is placed and a second region on which an edge ring is placed,

the electrode is a first electrode disposed below the first region,

the base further includes a second electrode,

the second electrode is disposed below the second region, forms the upper surface of the base, and is separated from the first electrode, and

the bias power supply or another bias power supply and another chuck power supply are electrically connected to the second electrode.

6. The plasma processing apparatus of claim 5, wherein the second electrode includes two electrodes constituting a bipolar electrode, and

said another chuck power supply includes two power supplies respectively connected to the two electrodes.

7. The plasma processing apparatus of claim 5, wherein the bias power supply or said another bias power supply is electrically connected to the second electrode via a capacitor and is separated from said another chuck power supply in a DC manner.

8. The plasma processing apparatus of claim 5, wherein the bias power supply or said another bias power supply and said another chuck power supply are electrically connected to the second electrode via a wiring extending between the upper surface and a bottom surface of the base member.

9. A plasma processing apparatus comprising:

a chamber;

a substrate support disposed in the chamber;

a plasma generator configured to generate plasma in the chamber;

a bias power supply configured to generate bias energy for attracting ions from the plasma to a substrate; and

a chuck power supply configured to generate a voltage applied to the substrate support to hold the substrate using an electrostatic attractive force,

16

wherein the bias energy is a radio frequency power or a periodically generated voltage pulse,

the substrate support includes:

a base; and

a dielectric part that provides a supporting surface on which a substrate is placed, extends from an upper surface of the base to the supporting surface, and is made of only a dielectric,

the supporting surface includes a first region on which the substrate is placed and a second region on which an edge ring is placed,

the base includes:

a first portion made of a conductive material and disposed below the first region; and

a second portion that is made of a conductive material, disposed below the second region, and separated from the first portion,

the bias power supply and the chuck power supply are electrically connected to the first portion, and

the bias power supply or another bias power supply and another chuck power supply are electrically connected to the second portion.

10. The plasma processing apparatus of claim 9, wherein the second portion includes two portions constituting a bipolar electrode, and

said another chuck power supply includes two power supplies respectively connected to the two portions.

11. The plasma processing apparatus of claim 9, wherein the bias power supply is electrically connected to the first portion via a capacitor and is separated from the chuck power supply in a DC manner, and

the bias power supply or said another bias power supply is electrically connected to the second portion via a capacitor and is separated from the separate chuck power supply in a DC manner.

12. The plasma processing apparatus of claim 9, wherein the plasma generator includes a radio frequency power supply electrically connected to the first portion of the base.

13. A substrate processing method using the plasma processing apparatus according to claim 1, comprising:

placing a substrate on the dielectric part of the substrate support;

generating plasma in the chamber using the plasma generator;

holding the substrate on the substrate support by supplying a voltage from the chuck power supply; and

attracting ions from the plasma to the substrate by supplying the bias energy from the bias power supply.

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